EEG-Based P300 Brain Speller

Report

1. Introduction

Brain-Computer Interfaces (BCIs) provide a way for humans to interact with external systems using only brain activity, without any physical movement. One of the most important applications of BCI is the P300 speller, which enables users to type characters using EEG signals.

The P300 speller works on the principle of event-related potentials (ERPs). When a user focuses on a specific character in a flashing matrix, the brain produces a P300 response approximately 300 ms after the target stimulus appears. By detecting this response, it is possible to identify the intended character.

The aim of this project is to implement a complete EEG signal processing pipeline to detect P300 responses and classify them using machine learning techniques.

2. Dataset and Tools

The dataset used in this project is the **BNCI2014-009 dataset**, which contains EEG recordings from subjects performing a P300 speller task. The dataset was accessed using the MOABB framework, which allows easy loading and handling of BCI datasets.

The implementation was carried out in Python using the following libraries:

- MNE for EEG signal processing
- MOABB for dataset handling
- NumPy and Pandas for data operations
- Scikit-learn for machine learning models
- Matplotlib and Seaborn for visualization

3. Methodology

The entire system was implemented as a sequence of well-defined steps, starting from raw EEG signals to final classification.

3.1 Preprocessing

EEG signals are usually noisy and contain unwanted components such as power-line interference and physiological artifacts. To improve signal quality, the following preprocessing steps were applied:

- A bandpass filter (0.1–30 Hz) was used to retain relevant EEG frequencies associated with P300.
- A notch filter at 50 Hz was applied to remove power-line noise.

- The EEG data was re-referenced using the average reference method.
- Independent Component Analysis (ICA) was used to remove artifacts such as eye blinks and muscle activity.

3.2 Epoching

The continuous EEG signal was segmented into smaller time windows (epochs) around each stimulus event. Each epoch represents the brain's response to a single flash.

- Time window: −200 ms to +800 ms
- Baseline correction was not applied, as it is not required when using Xdawn

3.3 Feature Extraction

To enhance the P300 signal, Xdawn spatial filtering was applied. Xdawn is specifically designed for ERP-based applications and helps improve the signal-to-noise ratio.

The transformed epochs were then flattened into feature vectors, which were used as input for the classifier.

3.4 Classification

A Support Vector Machine (SVM) with an RBF kernel was used for classification. Since the dataset is imbalanced (more non-target than target samples), class weighting was applied to ensure that the model pays sufficient attention to the P300 class.

Additionally, probability-based prediction with a lower threshold (0.3) was used to improve detection of P300 events.

3.5 Evaluation

The model was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score

A confusion matrix was also plotted to better understand the classification performance.

In addition, the Information Transfer Rate (ITR) was calculated to measure how efficiently the system can communicate information.

4. Results

The final model achieved the following results:

- Accuracy: approximately 84%
- Precision (P300 class): 0.52
- Recall (P300 class): 0.74
- F1-score (P300 class): 0.61

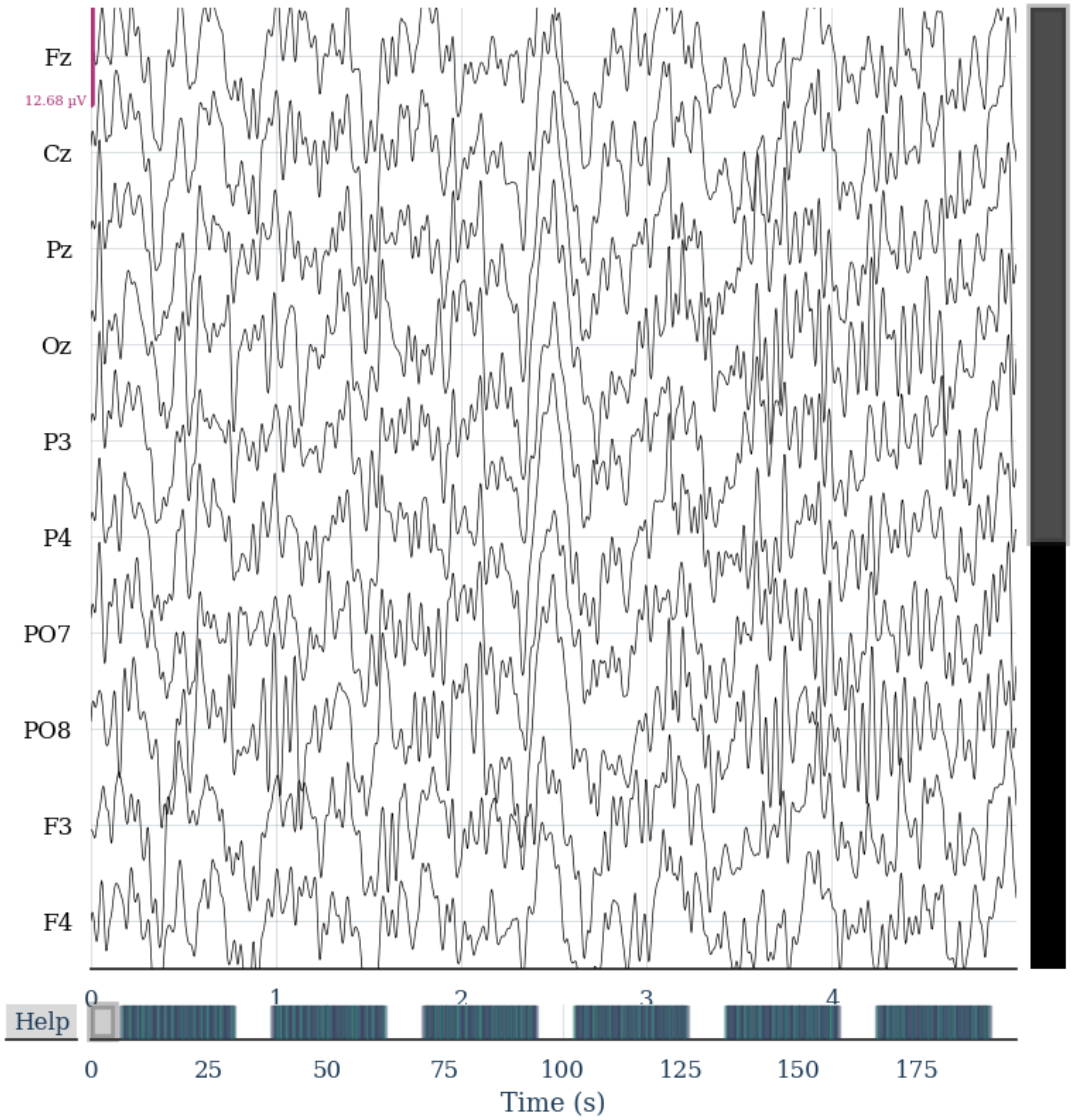
```
Accuracy: 0.8362068965517241
              precision    recall  f1-score   support

           0       0.94      0.86      0.90        97
           1       0.50      0.74      0.60        19

    accuracy                           0.84       116
   macro avg       0.72      0.80      0.75       116
weighted avg       0.87      0.84      0.85       116
```

The confusion matrix shows that the model performs very well in identifying non-target signals and reasonably well in detecting P300 targets.

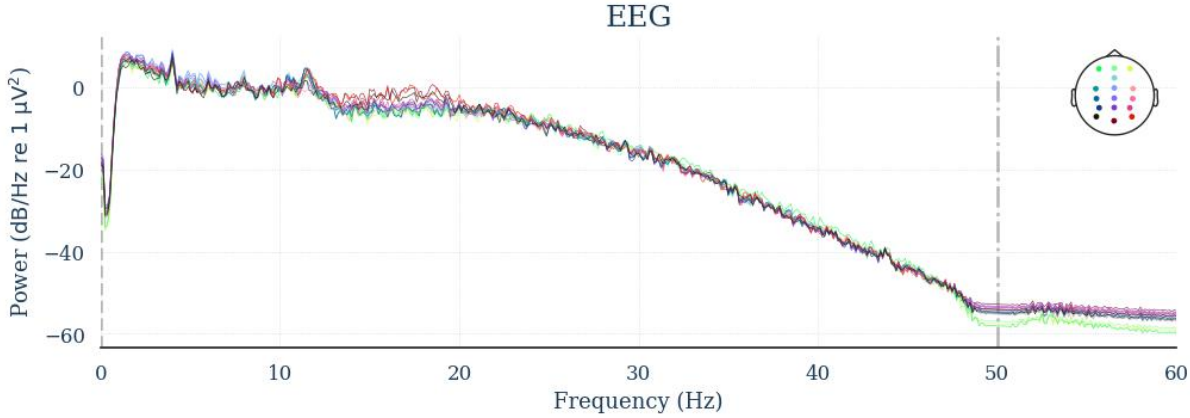
The Information Transfer Rate was calculated to be approximately **45 bits per minute**, assuming a realistic trial duration of 5 seconds.

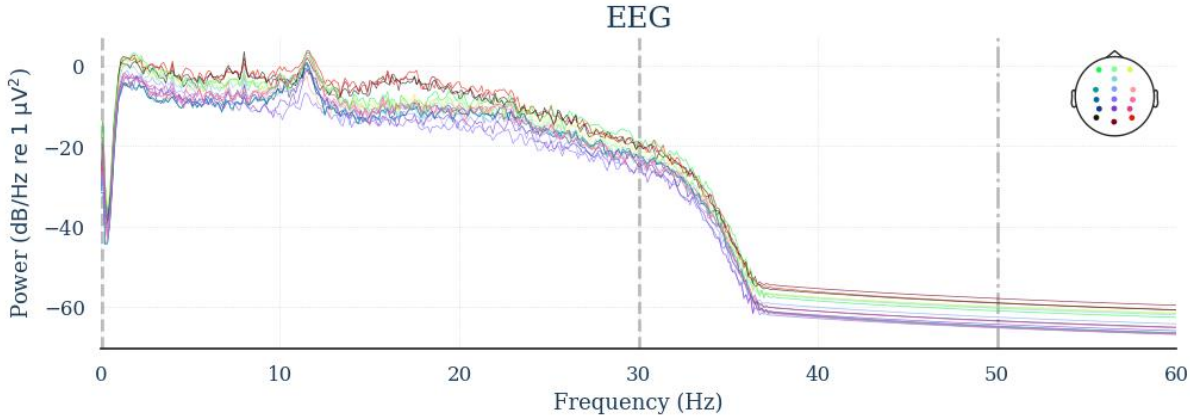
5. Plots:

- **Raw EEG Signal Plot**

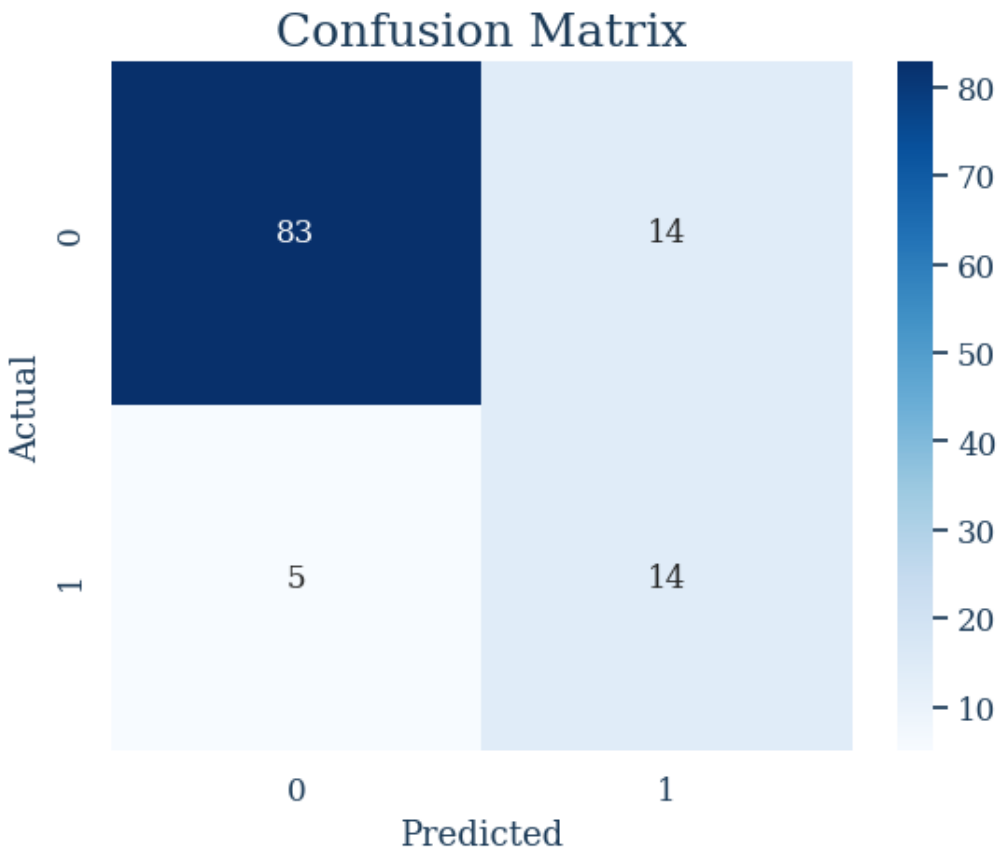
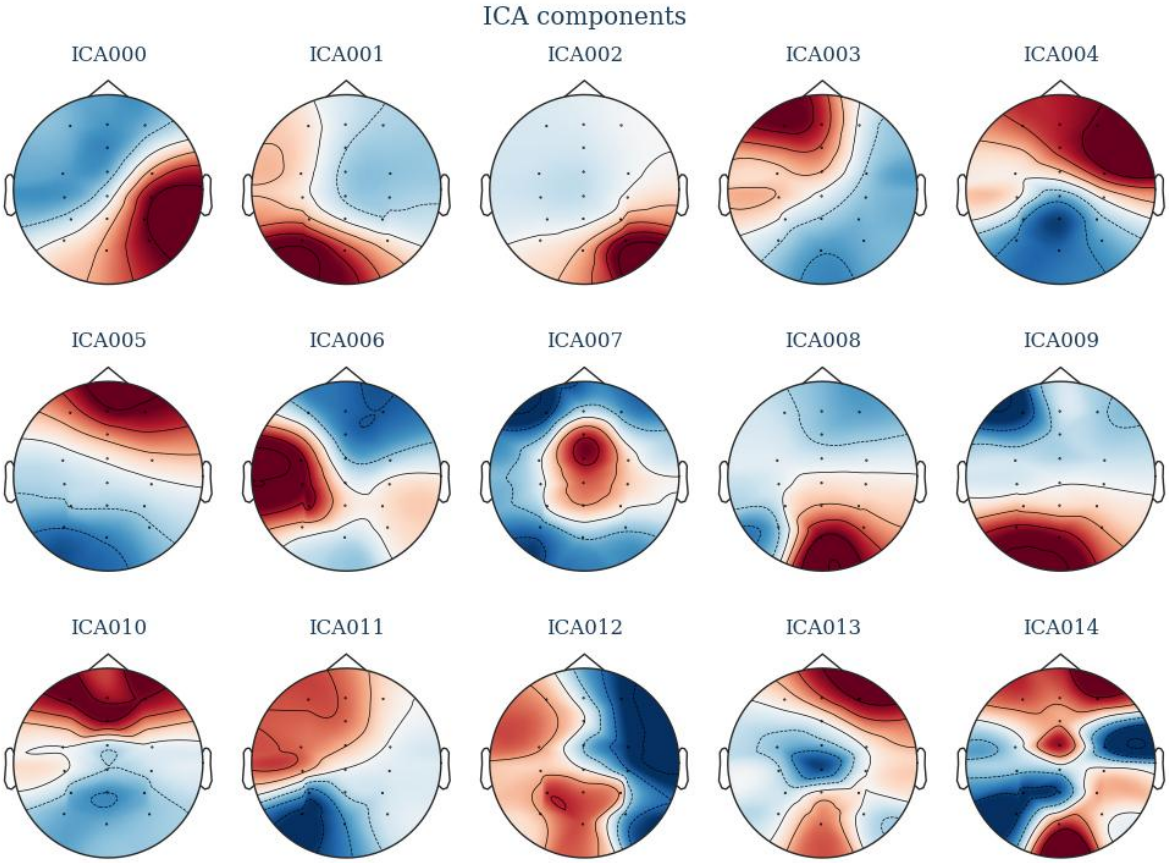


- **PSD before and after filtering**





- **Confusion Matrix**



- **ICA component plot**